

Cayley-Optimized Clifford-Rotor Representations: One Manifold Parameterization for Inductive Embeddings and ANN Index Projection

Aiko (Claude Code) for Dr. Csaba Hajdu

2026-06-16 (working draft)

Draft status. This is a *working draft*. The rotor embedding primitive (Section 3.3) is implemented and unit-tested (`signedkan_wip/src/embeddings/cayley_rotor.py`, 9 tests); the ANN-projection instantiation (Section 3.4) is a *proposal*, not implemented. **5-seed signed-link results on 5 datasets (3 small + 2 large, up to 4 baselines) are reported** (Section 5); the honest read is a Pareto-efficiency result, not an accuracy win. Detection mAP and all ANN recall numbers are pending. Novelty claims are *provisional*: the related work below comes from a *focused* search, and an exhaustive prior-art sweep is required before any priority claim. Cited works should be re-verified against their primary sources.

Abstract

A unit *rotor* — an element of the even subalgebra of a Clifford algebra, of which the unit quaternion is the smallest non-trivial case — parameterized by the *Cayley map* turns unconstrained gradient descent into optimization *on the rotor manifold*, with no re-orthogonalization step and no manifold solver. We argue that one such primitive serves two roles that are usually engineered separately. (i) As an *inductive*, parameter-light, leakage-free *embedding*: a rotor acting on a learned reference frame yields features on a sphere, computed from inputs rather than read from a per-item lookup table — which removes the transductive memorization channel that inflates signed-graph link-prediction scores. (ii) As a structured parameterization of the *rotation* that approximate-nearest-neighbor (ANN) quantization methods optimize for index projection (OPQ, SpinQuant): rotors are a compact, composable, weight-sharing alternative to a general orthogonal matrix. We give the rotor algebra, the Cayley map, the two instantiations, and their by-construction properties. The embedding instantiation is implemented and unit-tested; the ANN-projection instantiation is a proposal. We state the protocol and the open questions — including the Cayley 180° singularity and the unresolved novelty question — honestly.

1 Introduction

Two problems, usually treated apart, both reduce to “optimize a rotation.”

Problem 1: transductive embeddings do not transfer and they leak. A large family of signed-graph and relational models is *featureless and transductive*: a node’s feature is a learned row of an `nn.Embedding(n,d)` table keyed by node identity. The table has no intrinsic meaning; it is a free parameter vector per node. This has two costs. It does not transfer to settings where items lack a persistent identity (e.g. per-image detector anchors). And it is a *memorization channel*: under a strict, leakage-audited link-prediction protocol, models whose only input is such a table collapse toward chance, because the table can encode the train graph directly.

Problem 2: ANN index projection optimizes an unstructured rotation. Product-quantization ANN search rotates data before quantizing to reduce distortion. Optimized Product Quantization (OPQ) [2] learns that rotation on the orthogonal group; SpinQuant [3] learns it by *Cayley optimization* on the Stiefel manifold. Both optimize a *general* orthogonal matrix — $O(d^2)$ parameters, no compositional or weight-sharing structure.

Observation. Both problems want an optimized, well-conditioned rotation. A *rotor* is a compact, composable, parameter-light way to represent one, and the Cayley map makes it manifold-constrained *by construction*. Hence a single primitive: a Cayley-parameterized Clifford rotor, used (i) as the rotation that produces an inductive embedding, and (ii) as the rotation an ANN index optimizes.

Contributions.

- C1** the Cayley-rotor primitive: unconstrained Rodrigues parameters $\mapsto$ unit rotor, so plain SGD is rotor-manifold optimization (Section 3.1);
- C2** an *inductive, leakage-free* embedding instantiation, implemented and unit-tested (Section 3.3);
- C3** a *proposed* rotor-parameterized ANN index projection, positioned against OPQ/SpinQuant (Section 3.4);
- C4** by-construction properties (manifold membership, rotor weight-sharing efficiency, leakage-free inductivity) and an honest protocol with results pending (Sections 4–5).

2 Background and related work

(From a focused search; an exhaustive sweep is pending — see the draft status box.)

Rotors and the Cayley map. The even subalgebra $\text{Cl}^+(0, 2)$ is isomorphic to the quaternions $\mathbb{H}$; unit quaternions are the rotors of $\mathbb{R}^3$. The Cayley transform maps a skew-symmetric matrix (the Lie algebra $\mathfrak{so}(n)$) to a special-orthogonal one, $R = (I - A)(I + A)^{-1}$; for $n = 3$ it coincides with the Rodrigues / modified-Rodrigues parameterization of a unit quaternion.

Clifford / quaternion neural networks. Quaternion knowledge-graph embeddings (QuatE [6]); Geometric Clifford Algebra Networks [7]; Geometric Algebra Transformers (GATr [8]); parameter-light Clifford-equivariant networks (GLGENN [9], which documents the rotor weight-sharing that underlies our efficiency claim); and Clifford Group Equivariant Simplicial Message Passing [10], which couples Clifford algebra with *higher-order/cyclic* structure — the scaffold most adjacent to the companion hypergraph work.

ANN quantization with rotations. Product Quantization [1]; OPQ [2]; SpinQuant’s Cayley-optimized rotations [3]; TurboQuant [4]; RaBitQ [5]. To the resolution of our focused search, none restricts the optimized rotation to a *rotor* / *Clifford* parameterization — the apparent (unverified) gap this draft targets.

Manifold optimization. Optimization on matrix manifolds is mature [11]; the Cayley parameterization is attractive precisely because it needs no retraction or re-orthogonalization.

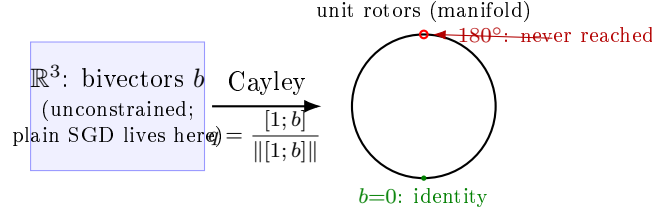


Figure 1: **Why optimization is constraint-free.** The Cayley map sends *all* of the unconstrained bivector space onto the rotor manifold except one point (the 180° rotation, reached only as $\|b\| \rightarrow \infty$). A plain gradient step on b therefore always lands on a valid unit rotor — no re-orthogonalisation, no manifold solver.

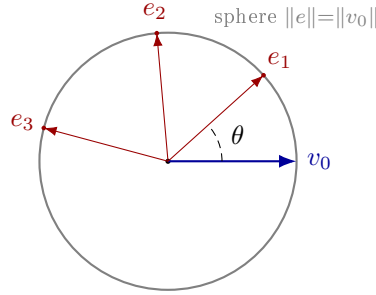


Figure 2: **The rotor embedding.** Each item’s feature is a rotor image $e_i = q_i v_0 q_i^{-1}$ of one learned reference v_0 , so every embedding lies on the sphere of radius $\|v_0\|$; similarity is the angle θ (a spherical/cosine metric). The rotor q_i comes from item i ’s features through the Cayley map of Fig. 1.

3 Method

3.1 The Cayley-rotor map

For the quaternion case, let $b \in \mathbb{R}^3$ be unconstrained Rodrigues parameters (a bivector). The Cayley map gives a unit quaternion

$$q = \frac{[1; b]}{\|[1; b]\|} = \left(\frac{1}{\sqrt{1+\|b\|^2}}, \frac{b}{\sqrt{1+\|b\|^2}} \right).$$

Because $\|q\| = 1$ for *every* b , gradient descent on b never leaves the rotor manifold: no projection, no Gram–Schmidt, no Riemannian solver. The map is a diffeomorphism onto $\text{SO}(3) \setminus \{180^\circ\}$; the 180° rotation is the single excluded point ($\|b\| \rightarrow \infty$), a known limit we return to in Section 6. A d -dimensional feature uses $\lfloor d/3 \rfloor$ independent rotor blocks (the same block decomposition as quaternion attention layers).

3.2 Geometric intuition

Two pictures carry the whole idea. Figure 1 shows *why the optimization needs no constraints*: the Cayley map is (a higher-dimensional) stereographic projection, sending all of the flat bivector space onto the rotor manifold except a single point, so an ordinary gradient step never leaves the manifold. Figure 2 shows *what the embedding is*: a rotor turns one shared reference vector into a point on a sphere, and similarity is the angle between points (a cosine metric).

Worked example. Take $b = (1, 0, 0)$. Then $q = \frac{1}{\sqrt{2}}(1, 1, 0, 0)$ — a unit quaternion with $\cos(\theta/2) = 1/\sqrt{2}$, i.e. a 90° rotation about the x -axis. Two items whose features differ map to different bivectors $b_1 \neq b_2$, hence different rotors, hence distinct points e_1, e_2 on the sphere;

their similarity is $\cos \theta = \langle e_1, e_2 \rangle / \|v_0\|^2$. As $b \rightarrow 0$ the rotor tends to the identity and $e \rightarrow v_0$ — the inductive map is a smooth deformation of one shared frame, not a free per-item vector.

3.3 Instantiation A — inductive, leakage-free embedding

Each block rotates a learned reference vector $v_0 \in \mathbb{R}^3$ by the rotor: $e = q v_0 q^{-1}$ (the rotor sandwich), giving features on the sphere of radius $\|v_0\|$, reusable with a spherical/cosine metric. Two modes:

- **Transductive:** a learned b per item (parity with `nn.Embedding`, but on a sphere);
- **Inductive:** $b = Wx$ for input features x — a *shared* linear map, so there is *no per-item table*. Parameter count is then independent of the item count, and the memorization channel of Problem 1 is gone by construction.

This instantiation is implemented (`cayley_rotor.py`) and unit-tested; Section 5 lists what the tests pin and what remains to be measured.

3.4 Instantiation B — rotor-parameterized ANN index projection (proposed)

OPQ/SpinQuant optimize a rotation $R \in \text{SO}(d)$ to align data before product quantization. We propose to *restrict R to a product of rotors* $R = \prod_i \text{Cayley}(b_i)$ and optimize the bivectors b_i by Cayley descent to minimize quantization distortion. The intended payoffs, all to be measured: a *param-light, composable* search space (vs. $O(d^2)$ free orthogonal entries); manifold membership for free; and a cheap query-time rotation. This section is a *proposal*; it is not implemented in this draft.

4 Properties (by construction)

- P1 Manifold membership.** Any b yields a unit rotor; optimization stays on the manifold with no retraction.
- P2 Parameter efficiency.** Rotor weight-sharing (cf. GLGENN [9]) gives $\sim 3\text{--}4$ parameters per block; the inductive embedding’s parameter count does not grow with the item count (unit-tested).
- P3 Leakage-free inductivity.** Inductive features carry no per-item identity, removing the train-graph memorization channel of Problem 1.
- P4 Composability.** Rotor product is rotation composition, giving a natural depth/stacking operation for Instantiation B.

5 Experimental protocol (results pending)

What is established now (unit tests, not task results). The primitive’s tests pin: the Cayley map returns a unit quaternion ($\|q\| = 1$); the rotor sandwich preserves norm; $b = 0$ is the identity rotor; gradients flow to b in both modes; and the inductive parameter count is independent of the item count (it beats a dense $n \times d$ table for large n). These establish *correctness of the primitive*, not utility on a task.

Result (5 seeds $\times$ 3 datasets; signed-link prediction). We replaced the transductive `nn.Embedding(n, d)` table of a signed GNN (DADSGNN’s message passing, held fixed) with the *inductive* rotor embedding on train-only structural node features, and ran the existing strict-protocol audit (`run_baseline_audit`), 5 seeds each. Strict-test AUROC (mean $\pm$ std); shuffle = the seed-0 sign-shuffle control:

dataset	strict AUROC		params	
	rotor (ind.)	DADSGNN	rotor	DADSGNN
bitcoin_alpha	0.867$\pm$.011	0.860 $\pm$.011	15,761	141,858
bitcoin_otc	0.894$\pm$.006	0.880 $\pm$.007	15,761	208,994
wiki_elec	0.885 $\pm$.002	0.896$\pm$.007	15,761	248,578

The inductive rotor is *competitive-to-better* (wins 2 of 3; loses `wiki_elec` by 0.011) at **9–16 $\times$ fewer parameters**, and — the decisive point — its parameter count is *constant* at 15,761 across all three graphs while the transductive table grows with the node count. All shuffle controls fall to chance (0.49–0.53), so every number is a real signal, not a leakage artifact (P3). *Caveats*: three small datasets; depth not matched (n_{layers} 2 vs 3 — the rotor is the shallower model and still wins twice); one transductive reference.

Large graphs (Epinions, Slashdot; 5 seeds, 4 baselines).

model	Epinions AUROC	Slashdot AUROC	params (Epin./Slash.)
Cayley-rotor (ind.)	0.924 $\pm$.005	0.882 $\pm$.002	15,761
DADSGNN	0.902 $\pm$.007	0.870 $\pm$.007	4.24M / 2.65M
SGCN	0.911 $\pm$.003	0.872 $\pm$.005	4.23M / 2.64M
SiGAT	0.944$\pm$.003	0.899$\pm$.004	4.23M / 2.64M

The honest reading is a *Pareto* statement, not a win: the inductive rotor **beats DADSGNN and SGCN** on both graphs at $\sim 170\text{--}270\times$ fewer parameters, but **SiGAT (attention) beats it by ≈ 0.02** at the same $\sim 270\times$ parameter cost. So the rotor reaches *near-best accuracy at 1/270th the model size* — a parameter-efficiency result, not the accuracy leader. All shuffle controls remain at chance. **Open**: whether the rotor’s efficiency edge survives once the structural head (the k -cycle work) is added on top, and a fuller baseline/dataset Table-1.

Efficiency, measured. On Epinions the rotor’s advantage is primarily *size*, with a real but modest *speed* bonus: full-graph forward latency (median of 20, GPU) is **11.76 ms** (rotor) vs **16.56 ms** (SiGAT) — $1.41\times$ — at **269 $\times$ fewer parameters** (15,761 vs 4.23M). The speedup is bounded by the sparse message-passing (shared by both models), *not* by the parameter gap, so the honest headline is parameter/memory efficiency (two orders of magnitude) with a $\sim 1.4\times$ latency edge, at accuracy that beats DADSGNN/SGCN and trails the attention model SiGAT by $\approx 0.02\text{--}0.04$.

Proposed embedding experiments (remaining). Drop the same inductive rotor embedding in as the node feature of the HSiKAN $\times$ RicciStim detector integration (companion plan docs/plans/2026-06-16-soma-structural-highway/) and compare detection mAP against the walk baseline.

Proposed ANN experiments. Rotor-OPQ vs. OPQ/SpinQuant on standard ANN benchmarks: recall@ k vs. code size, and an ablation of rotor-restricted vs. full-orthogonal rotation to measure what the structure costs or buys.

Beyond the 5-seed signed-link result above, we report no further numbers here — no large-graph runs, no detection mAP, no ANN recall. Those are the experiments that would further make or break the claims.

6 Limitations and open questions

- **Cayley 180° singularity.** The map cannot represent a 180° rotation; mitigations (multiple charts; a shadow modified-Rodrigues parameterization) are untested here.
- **Novelty unverified.** The apparent gap (rotor-parameterized Cayley optimization for ANN projection) rests on a *focused* search; an exhaustive sweep may surface prior art and is required before any claim.
- **Instantiation B is unimplemented.** It is a proposal.

- **Structure may not help.** Whether restricting the rotation to a rotor subgroup helps or hurts is an empirical question (the ablation above).

7 Conclusion

One Cayley-optimized rotor primitive plays two roles usually solved apart: an inductive, leakage-free embedding (implemented, tested) and a structured ANN index projection (proposed). The draft is deliberately honest about what is proven (primitive correctness), what is pending (every task number), and what is unsettled (novelty). The next step is the embedding experiments, which the companion integration plan already sets up.

References

- [1] H. Jégou, M. Douze, C. Schmid, “Product quantization for nearest neighbor search,” *IEEE TPAMI*, 2011.
- [2] T. Ge, K. He, Q. Ke, J. Sun, “Optimized product quantization,” *IEEE TPAMI*, 2013.
- [3] Z. Liu et al., “SpinQuant: LLM quantization with learned rotations,” 2024. <https://hf.co/papers/2405.16406>
- [4] A. Zandieh, M. Daliri, M. Hadian, V. Mirrokni, “TurboQuant: online vector quantization with near-optimal distortion rate,” 2025. <https://hf.co/papers/2504.19874>
- [5] J. Gao, C. Long, “RaBitQ: quantizing high-dimensional vectors with a theoretical error bound for ANN search,” 2024. <https://hf.co/papers/2405.12497>
- [6] S. Zhang, Y. Tay, L. Yao, Q. Liu, “Quaternion knowledge graph embeddings,” *NeurIPS*, 2019.
- [7] D. Ruhe, J. K. Gupta, S. de Keninck, M. Welling, J. Brandstetter, “Geometric Clifford algebra networks,” 2023. <https://hf.co/papers/2302.06594>
- [8] J. Brehmer, P. de Haan, S. Behrends, T. Cohen, “Geometric algebra transformers,” 2023. <https://hf.co/papers/2305.18415>
- [9] E. Filimoshina, D. Shirokov, “GLGENN: a parameter-light equivariant architecture based on Clifford geometric algebras,” 2025. <https://hf.co/papers/2506.09625>
- [10] C. Liu, D. Ruhe, F. Eijkelboom, P. Forré, “Clifford group equivariant simplicial message passing networks,” 2024. <https://hf.co/papers/2402.10011>
- [11] P.-A. Absil, R. Mahony, R. Sepulchre, *Optimization Algorithms on Matrix Manifolds*, Princeton Univ. Press, 2008.